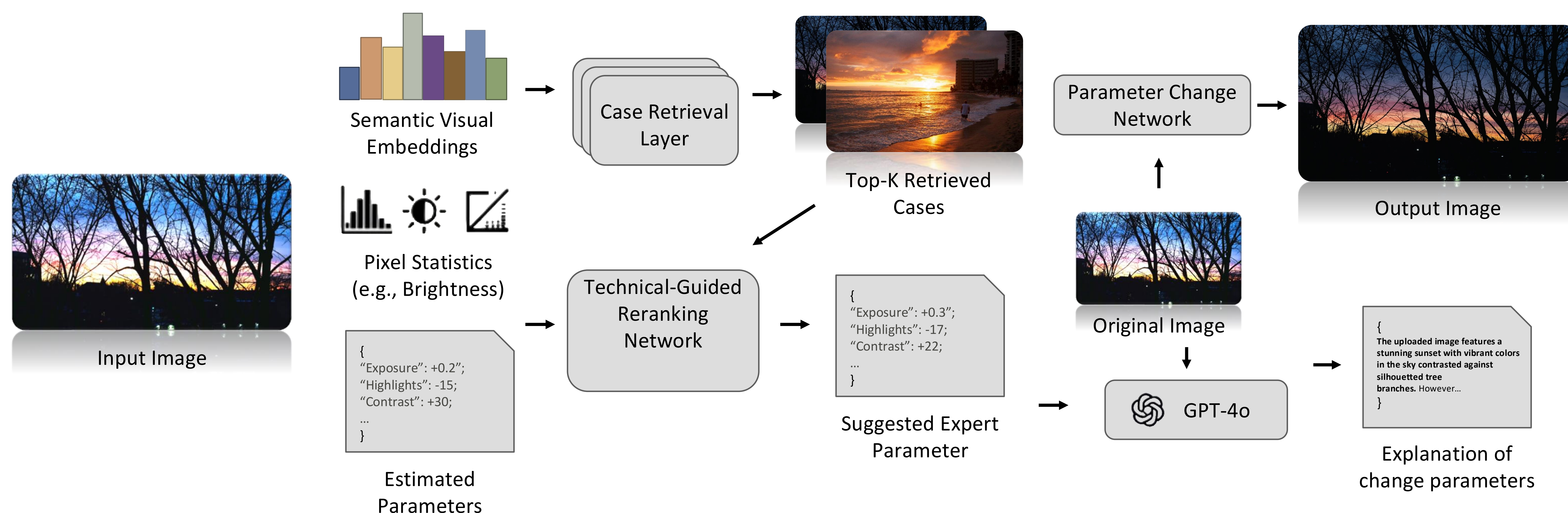




**Overview:** Achieving professional-grade photo enhancement requires translating complex visual context into precise UI parameters—a task where rigid auto-enhancers and generic Vision-Language Models (VLMs) often fail or hallucinate. LumiGrade AI solves this by introducing a Multi-Modal RAG architecture that grounds VLM reasoning in indexed expert photography knowledge. By mapping images into semantic and statistical representations, our system retrieves contextually matched expert recipes and reasons over them to generate safe, explainable, and image-specific color grading adjustments.

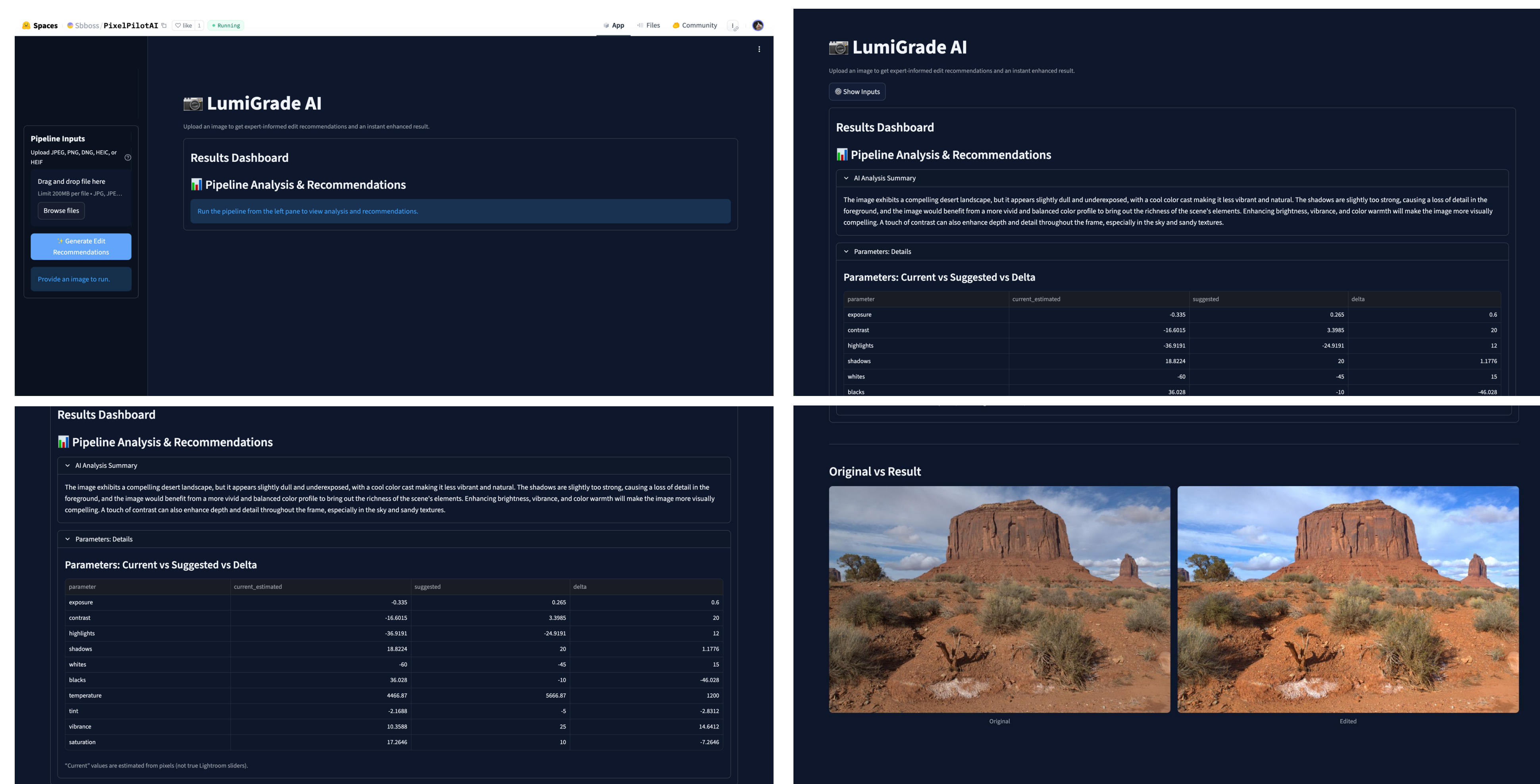
## Architecture Pipeline Overview



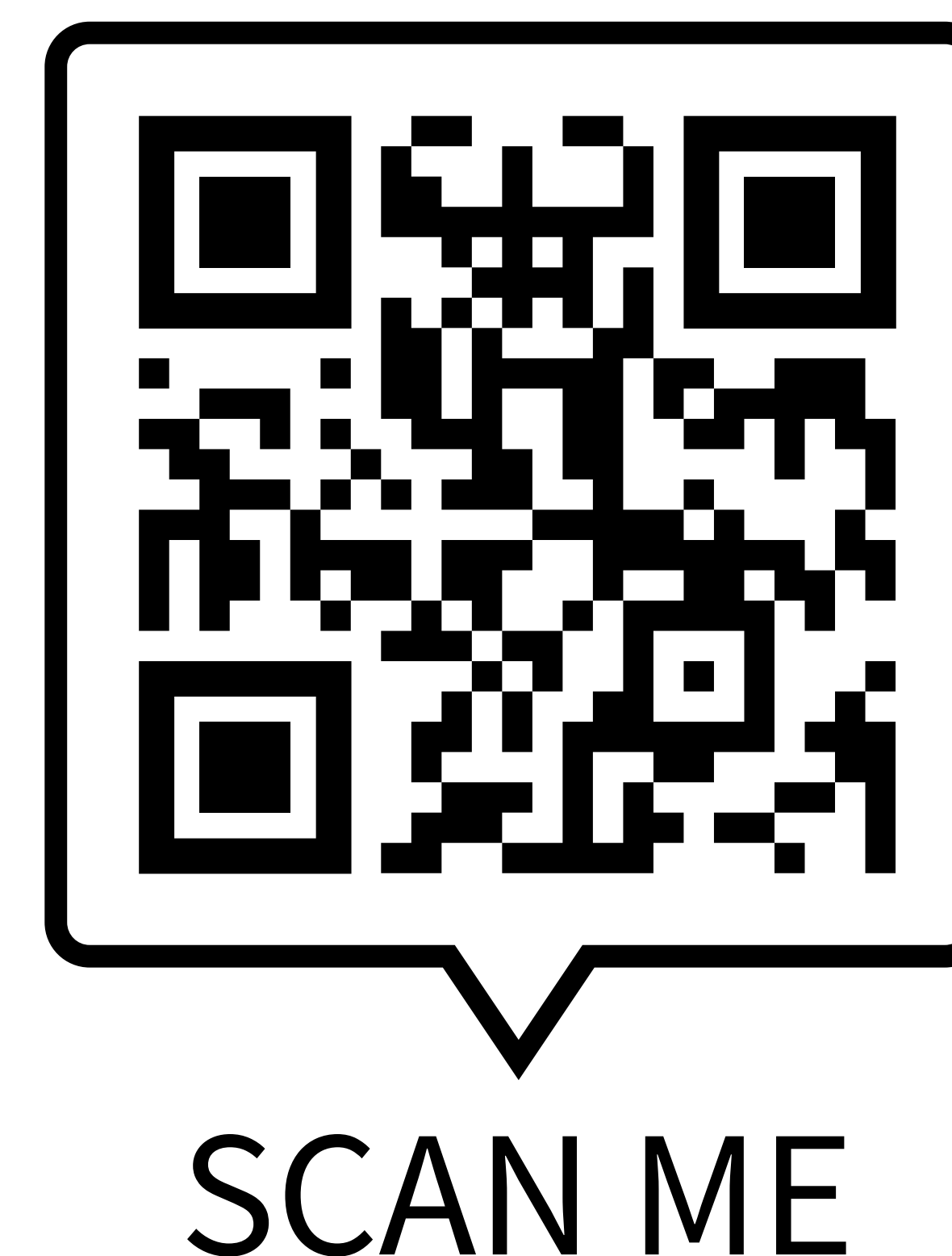
## Solution

- **Dual Representation:** Extract semantic embeddings (CLIP) and objective pixel statistics (e.g., clipping limits) from the input image.
- **Case-Based Retrieval:** Query a vector database using these representations to find lighting-matched, expert-level editing recipes.
- **Multi-Modal Reasoning:** A Vision-Language Model (GPT-4o) jointly analyzes the visual input and retrieved expert baselines to synthesize optimal parameter shifts.
- **Programmatic Guardrails:** Apply mathematical constraints to the LLM's output to prevent destructive edits, safely rendering the final image alongside an explainable summary.

## LumiGrade AI Website Preview



## Experience the APP



## References

[1] You Z, Wang K, Zhang H, et al. PhotoFramer: Multi-modal Image Composition Instruction[J]. arXiv preprint arXiv:2512.00993, 2025.

[2] Chen H, Tao K, Wang Y, et al. PhotoArtAgent: Intelligent Photo Retouching with Language Model-Based Artist Agents[J]. arXiv preprint arXiv:2505.23130, 2025.